THE D-TH FITTING IDEAL OF THE DIFFERENTIALS CUTS OUT THE NON-SMOOTH LOCUS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file divisors.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Carrier window: statement and proof both lie inside the window 0-2000 granted to this lane by CONTROL 8.83.

Editorial note. The two references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: sources/stacks-divisors/retained-preamble.tex (88 lines, verbatim present).

Allowed standard prerequisites (editorial).

- The rank strata of a module and their relation to Fitting ideals. Lemma lemma-locally-free-rank-r-pullback (1383-1422), used at proof lines 1628-1630. Describes the complement of Z as the locus where the differentials need at most d generators.
- Smoothness at a point via generators of the differentials. Morphisms, Lemma morphisms-lemma-smooth-at-point, used at proof lines 1630-1632. Identifies that locus with the non-smooth locus under the stated hypotheses.

Source reference. Original lemma 05P8: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7332–7332. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 01V9: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2438–2438. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let $f : X \to S$ be a morphism of schemes. Let $d \geq 0$ be an integer. Assume*

(1) *f is flat,*
(2) *f is locally of finite presentation, and*
(3) *every nonempty fibre of f is equidimensional of dimension d.*

Let $Z \subset X$ be the closed subscheme cut out by the dth fitting ideal of $\Omega_{X/S}$. Then Z is exactly the set of points where f is not smooth.

Proof. By Lemma 05P8 the complement of Z is exactly the locus where $\Omega_{X/S}$ can be generated by at most d elements. Hence the lemma follows from Morphisms, Lemma 01V9. □

Date: stacks-project:divisors.tex:local-snapshot-20261012; GNU FDL 1.2 or later.